

EEE4114F: Problem set 3 solutions

1.

The z -transform of a sequence $x(n)$ is

$$X(z) = \frac{z + 2z^{-2} + z^{-3}}{1 - 3z^{-4} + z^{-5}}$$

If the region of convergence includes the unit circle, find the DTFT of $x(n)$ at $\omega = \pi$.

If $X(z)$ is the z -transform of $x(n)$, and the unit circle is within the region of convergence, the DTFT of $x(n)$ may be found by evaluating $X(z)$ around the unit circle:

$$X(e^{j\omega}) = X(z)|_{z=e^{j\omega}}$$

Therefore, the DTFT at $\omega = \pi$ is

$$X(e^{j\pi})|_{\omega=\pi} = X(z)|_{z=e^{j\pi}} = X(z)|_{z=-1}$$

and we have

$$X(e^{j\pi})|_{\omega=\pi} = \frac{z + 2z^{-2} + z^{-3}}{1 - 3z^{-4} + z^{-5}} \bigg|_{z=-1} = \frac{-1 + 2 - 1}{1 - 3 - 1} = 0$$

2.

(a) Because this sequence is finite in length, the z -transform is a polynomial,

$$X(z) = 3 + z^{-2} + z^2$$

and the region of convergence is $0 < |z| < \infty$. Note that because $x(n)$ has nonzero values for $n < 0$, the region of convergence does not include $|z| = \infty$, and because $x(n)$ has nonzero values for $n > 0$, the region of convergence does not include the point $z = 0$.

(b) For this sequence,

$$X(z) = \sum_{n=0}^9 z^{-n} = \frac{1 - z^{-10}}{1 - z^{-1}}$$

which converges for all $|z| > 0$. Note that the roots of the numerator are solutions to the equation

$$z^{10} = 1$$

These roots are

$$z = e^{j2\pi k/10} \quad k = 0, 1, \dots, 9$$

which are 10 equally spaced points around the unit circle. Thus, the pole at $z = 1$ in the denominator of $X(z)$ is canceled by the zero at $z = 1$ in the numerator, and the z -transform may also be expressed in the form

$$X(z) = \prod_{k=1}^9 (1 - e^{jk\frac{2\pi}{10}} z^{-1})$$

3.

(a) Using the definition of the z -transform we have

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} = \sum_{n=-\infty}^0 \left(\frac{1}{3}\right)^n z^{-n}$$

where the sum converges for

$$|3z| < 1 \quad \text{or} \quad |z| < \frac{1}{3}$$

Alternatively, note that the time-reversed sequence $y(n) = x(-n) = \left(\frac{1}{3}\right)^{-n} u(n)$ has a z -transform given by

$$Y(z) = \frac{1}{1 - 3z^{-1}}$$

with a region of convergence given by $|z| > 3$. Therefore, using the time-reversal property, $Y(z) = X(z^{-1})$, we obtain the same result.

(b) Because $x(n)$ is the sum of two sequences, we will find the z -transform of $x(n)$ by finding the z -transforms of each of these sequences and adding them together. The z -transform of the first sequence may be found easily using the shift property. Specifically, note that because

$$\left(\frac{1}{3}\right)^n u(n+2) = 4\left(\frac{1}{3}\right)^{n+2} u(n+2)$$

the z -transform of $\left(\frac{1}{3}\right)^n u(n+2)$ is $4z^2$ times the z -transform of $\left(\frac{1}{3}\right)^n u(n)$, that is,

$$\left(\frac{1}{3}\right)^n u(n+2) \xleftrightarrow{Z} \frac{4z^2}{1 - \frac{1}{3}z^{-1}}$$

which has a region of convergence $|z| > \frac{1}{3}$.

The second term is a left-sided exponential and has a z -transform that we have seen before, that is,

$$3^n u(-n-1) \xleftrightarrow{Z} -\frac{1}{1 - 3z^{-1}}$$

with a region of convergence $|z| < 3$.

Finally, for the z -transform of $x(n)$, we have

$$X(z) = \frac{4z^2}{1 - \frac{1}{3}z^{-1}} - \frac{1}{1 - 3z^{-1}}$$

with a region of convergence $\frac{1}{3} < |z| < 3$.

(c) As we saw in Problem 4.3(b), the z -transform of $\cos(n\omega_0)u(n)$ is

$$\cos(n\omega_0)u(n) \xleftrightarrow{Z} \frac{1 - (\cos \omega_0)z^{-1}}{1 - 2(\cos \omega_0)z^{-1} + z^{-2}}; \quad |z| > 1$$

Therefore, using the exponentiation property,

$$\alpha^n x(n) \xleftrightarrow{Z} X(\alpha^{-1}z)$$

we have

$$\left(\frac{1}{3}\right)^n \cos(n\omega_0)u(n) \xleftrightarrow{Z} \frac{1 - \frac{1}{3}(\cos \omega_0)z^{-1}}{1 - \frac{2}{3}(\cos \omega_0)z^{-1} + \frac{1}{9}z^{-2}}$$

with a region of convergence $|z| > \frac{1}{3}$.

(d) Writing $x(n)$ as

$$x(n) = \alpha^n u(n) + \alpha^{-n} u(-n) - \delta(n)$$

we may use the linearity and time-reversal properties to write

$$X(z) = \frac{1}{1 - \alpha z^{-1}} + \frac{1}{1 - \alpha z} - 1 \quad \frac{1}{3} < |z| < 2$$

which may be simplified to

$$X(z) = \frac{1 - \alpha^2}{(1 - \alpha z^{-1})(1 - \alpha z)} \quad \frac{1}{3} < |z| < 2$$

4.

(a) The derivative property states that if $X(z)$ is the z -transform of $x(n)$,

$$nx(n) \xleftrightarrow{z} -z \frac{d}{dz} X(z)$$

If we let $x(n) = nw(n)$, where

$$w(n) = \left(\frac{1}{2}\right)^n u(n-2) = \frac{1}{4} \left(\frac{1}{2}\right)^{n-2} u(n-2)$$

from the delay property and the z -transform pair

$$\alpha^n u(n) \xleftrightarrow{z} \frac{1}{1 - \alpha z^{-1}} \quad |z| > |\alpha|$$

it follows that

$$W(z) = \frac{\frac{1}{4} z^{-2}}{1 - \frac{1}{2} z^{-1}} \quad |z| > \frac{1}{2}$$

Therefore, using the derivative property, we have the z -transform of $x(n)$,

$$X(z) = -z \frac{d}{dz} W(z) = \frac{1}{2} z^{-2} \frac{1 - \frac{1}{4} z^{-1}}{\left(1 - \frac{1}{2} z^{-1}\right)^2}$$

(b) Evaluating the z -transform of this sequence directly is difficult due to the factor of n^{-1} . However, if we define a new sequence, $y(n)$, as follows,

$$y(n) = nx(n) = (-2)^{-n} u(-n-1)$$

the z -transform of $y(n)$ is easily determined to be

$$Y(z) = \frac{-1}{1 + \frac{1}{2} z^{-1}} \quad |z| < \frac{1}{2}$$

Noting the relationship between $x(n)$ and $y(n)$, we can apply the derivative property to set up a differential equation for $X(z)$,

$$-z \frac{d}{dz} X(z) = \frac{-1}{1 + \frac{1}{2} z^{-1}}$$

or

$$\frac{d}{dz} X(z) = \frac{1}{z + \frac{1}{2}}$$

The solution to this differential equation is

$$X(z) = \log \left(z + \frac{1}{2} \right)$$

and the region of convergence is $|z| < \frac{1}{2}$.

5.

To evaluate this convolution, we will use the convolution property of the z -transform. The z -transform of $h(n)$ is

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} \quad |z| > \frac{1}{2}$$

and the z -transform of $x(n)$ may be found from the time-reversal and shift properties, or directly as follows:

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} x(n)z^{-n} = \sum_{n=-\infty}^0 3^n z^{-n} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{3}z\right)^n = \frac{1}{1 - \frac{1}{3}z} = -\frac{3z^{-1}}{1 - 3z^{-1}} \quad |z| < 3 \end{aligned}$$

Therefore, the z -transform of the convolution, $y(n) = x(n) * h(n)$, is

$$Y(z) = -\frac{1}{1 - \frac{1}{2}z^{-1}} \cdot \frac{3z^{-1}}{1 - 3z^{-1}}$$

The region of convergence is the intersection of the regions $|z| > \frac{1}{2}$ and $|z| < 3$, which is $\frac{1}{2} < |z| < 3$. To find the inverse z -transform, we perform a partial fraction expansion of $Y(z)$,

$$Y(z) = \frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{1 - 3z^{-1}}$$

where

$$A = \left[\left(1 - \frac{1}{2}z^{-1}\right) Y(z) \right]_{z=\frac{1}{2}} = \frac{6}{5}$$

and

$$B = \left[(1 - 3z^{-1}) Y(z) \right]_{z=3} = -\frac{6}{5}$$

Therefore, it follows that

$$y(n) = \left(\frac{6}{5}\right)\left(\frac{1}{2}\right)^n u(n) + \left(-\frac{6}{5}\right)3^n u(-n-1)$$

6.

$$\left(\frac{1}{2}\right)^n u(n) \xleftrightarrow{z} \frac{1}{1 - \frac{1}{2}z^{-1}}$$

it follows that the z-transform of $w(n) = n\left(\frac{1}{2}\right)^n u(n)$ is

$$W(z) = -z \frac{d}{dz} \frac{1}{1 - \frac{1}{2}z^{-1}} = \frac{\frac{1}{2}z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)^2}$$

Because $x(n)$ may be written as

$$x(n) = |n|\left(\frac{1}{2}\right)^{|n|} = n\left(\frac{1}{2}\right)^n u(n) - n\left(\frac{1}{2}\right)^{-n} u(-n)$$

using linearity and the time-reversal property, we have

$$X(z) = \frac{\frac{1}{2}z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)^2} + \frac{\frac{1}{2}z}{\left(1 - \frac{1}{2}z\right)^2} = \frac{\frac{5}{8}z + \frac{5}{8}z^{-1} - 1}{\left(1 - \frac{1}{2}z^{-1}\right)^2 \left(1 - \frac{1}{2}z\right)^2}$$

which has a region of convergence $\frac{1}{2} < |z| < 2$.

7. For a left-sided sequence that is zero for $n > 0$, the z-transform is

$$X(z) = x(0) + x(-1)z + x(-2)z^2 + \dots$$

Therefore, it follows that

$$x(0) = \lim_{z \rightarrow 0} X(z)$$

For the given z-transform, we see that

$$x(0) = \lim_{z \rightarrow 0} X(z) = \lim_{z \rightarrow 0} \frac{3z^{-1} + 2z^{-2}}{3 - z^{-1} + z^{-2}} = \lim_{z \rightarrow 0} \frac{3z + 2}{3z^2 - z + 1} = 2$$

8.

- (a) Because $X(z)$ is a finite-order polynomial, $x(n)$ is a finite-length sequence. Therefore, $x(n)$ is the coefficient that multiplies z^{-n} in $X(z)$. Thus, $x(0) = 4$ and $x(2) = x(-2) = 3$.
- (b) This z -transform is a sum of two first-order rational functions of z . Because the region of convergence of $X(z)$ is the exterior of a circle, $x(n)$ is a right-sided sequence. Using the z -transform pair for a right-sided exponential, we may invert $X(z)$ easily as follows:

$$x(n) = \left(\frac{1}{2}\right)^n u(n) + 3\left(\frac{1}{3}\right)^n u(n)$$

- (c) Here we have a rational function of z with a denominator that is a quadratic in z . Before we can find the inverse z -transform, we need to factor the denominator and perform a partial fraction expansion:

$$\begin{aligned} X(z) &= \frac{1}{1 + 3z^{-1} + 2z^{-2}} = \frac{1}{(1 + 2z^{-1})(1 + z^{-1})} \\ &= \frac{2}{1 + 2z^{-1}} - \frac{1}{1 + z^{-1}} \end{aligned}$$

Because $x(n)$ is right-sided, the inverse z -transform is

$$x(n) = 2(-2)^n u(n) - (-1)^n u(n)$$

- (d) One way to invert this z -transform is to perform a partial fraction expansion. With

$$\begin{aligned} X(z) &= \frac{1}{(1 - z^{-1})(1 - z^{-2})} = \frac{1}{(1 - z^{-1})^2(1 + z^{-1})} \\ &= \frac{A}{1 + z^{-1}} + \frac{B_1}{1 - z^{-1}} + \frac{B_2}{(1 - z^{-1})^2} \end{aligned}$$

the constants A , B_1 , and B_2 are as follows:

$$\begin{aligned} A &= [(1 + z^{-1})X(z)]_{z=-1} = \frac{1}{4} \\ B_1 &= \left[\frac{d}{dz} (1 - z^{-1})^2 X(z) \right]_{z=1} = \left[\frac{z^{-2}}{(1 + z^{-1})^2} \right]_{z=1} = \frac{1}{4} \\ B_2 &= [(1 - z^{-1})^2 X(z)]_{z=1} = \frac{1}{2} \end{aligned}$$

Inverse transforming each term, we have

$$x(n) = \frac{1}{4}[(-1)^n + 1 + 2(n + 1)]u(n)$$

Another way to invert this z -transform is to note that $x(n)$ is the convolution of the two sequences,

$$x(n) = x_1(n) * x_2(n)$$

where $x_1(n) = u(n)$ and $x_2(n)$ is a step function that is up-sampled by a factor of 2. Because

$$x_1(n) * x_2(n) = \{1, 1, 2, 2, 3, 3, 4, 4, \dots\}$$

we have the same result as before.

9.

$$X(z) = \frac{1 + \frac{1}{2}z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)^2} \quad |z| > \frac{1}{2}$$

Here we have a second-order pole at $z = \frac{1}{2}$. The partial fraction expansion for $X(z)$ is

$$X(z) = \frac{A_1}{1 - \frac{1}{2}z^{-1}} + \frac{A_2}{\left(1 - \frac{1}{2}z^{-1}\right)^2}$$

The constant A_1 is

$$A_1 = \frac{1}{2} \left[\frac{d}{dz} \left(1 - \frac{1}{2}z^{-1}\right)^2 X(z) \right]_{z=1/2} = \frac{1}{2} \left[-\frac{1}{2}z^{-2} \right]_{z=1/2} = -\frac{1}{2}$$

and the constant A_2 is

$$A_2 = \left[\left(1 - \frac{1}{2}z^{-1}\right)^2 X(z) \right]_{z=1/2} = \frac{1}{2}$$

Therefore,

$$X(z) = -\frac{\frac{1}{2}}{1 - \frac{1}{2}z^{-1}} + \frac{\frac{1}{2}}{\left(1 - \frac{1}{2}z^{-1}\right)^2}$$

and

$$x(n) = -\left(\frac{1}{2}\right)^{n+1} u(n) + 3(n+1)\left(\frac{1}{2}\right)^{n+1} u(n)$$

10. First we need to find $X(z)$ and $H(z)$

$$X(z) = 1 - 2z^{-2}$$

$$H(z) = 2 - 2z^{-1} + 3z^{-2} + z^{-3}$$

And because $Y(z) = H(z)X(z)$:

$$\begin{aligned} Y(z) &= (2 - 2z^{-1} + 3z^{-2} + z^{-3})(1 - 2z^{-2}) \\ &= 2 - 2z^{-1} - z^{-2} + 5z^{-3} - 6z^{-4} - 2z^{-5} \end{aligned}$$

Taking the inverse Z-transform,

$$2\delta[n] - 2\delta[n-1] - \delta[n-2] + 5\delta[n-3] - 6\delta[n-4] - 2\delta[n-5]$$